

Joshua Himmens

joshua@himmens.com | 587-434-0118 | himmens.com

Undergraduate Engineering Physics Student at The University of British Columbia

92% (A+) Average | **English, French** (Working Knowledge) | [Github/Joshuah143](https://github.com/Joshuah143)

Technical Skills

Machine Learning | TensorFlow, Keras, PointNet, Weights and Biases (wandb), ONNX

Embedded Programming | Experienced with FreeRTOS on TMS570, RP2040, and STM32

Software Development | C, C++, Python, Java, MATLAB, Bash, Git

Particle Physics | Root, CERN grid compute, Athena

Technical/Research Experience

Associate Machine Learning Developer

AltaML | 01/2025 - 04/2025

ATLAS Deep Learning Research Student

TRIUMF | 05/2024 - Present (part time from 09/2024) | himmens.com/triumf

- Developed panoptic segmentation models for the ATLAS detector using the PointNet ML framework with Wandb, TensorFlow, Keras.
- Used CERN's grid computing to parallelize compute across thousands of nodes.
- Worked independently to develop models using cutting edge transfer learning approaches.
- Implementing models into production using NVIDIA Triton.

Command and Data Handling (CDH) Lead and Firmware Developer

UBC Orbit Satellite Team | 10/2023 - Present | himmens.com/orbit

- Led the CDH team to develop software to meet mission and testing objectives from ESA (European Space Agency) for the ALEASAT project.
- Managed a team of 10 firmware developers, with over 40 tasks completed, 1300 CI builds, 2000 lines of code completed/written.
- Developed mission testing, function testing, and acceptance testing procedures to meet ESA and ECSS standards.
- Programmed device drivers and electrical ground support equipment (EGSE).

Publications and Presentations

Presented **Developing Machine Learning Techniques for Particle Flow in the ATLAS Experiment** at the Canadian Astroparticle Physics Summer Student Talks Competition (CASST 2024), where I **placed 2nd** of 44 presentations.

Presenter of **JetPointnet: A Machine Learning Approach to Cell-to-Track Attribution in the ATLAS Experiment** presented at the Canadian Undergraduate Physics Conference in October 2024.

Presented **3D Particle Flow in the ATLAS Calorimeter: How to Train Your Model**, a speed-talk, at the 2024 TRIUMF Science Week

Presented **ALEASAT ESA "Fly Your Satellite!" Training Week Presentation** at the European Space Agency's ESEC-GALAXIA (Transinne Belgium) in 2024 as part of the "Fly Your Satellite 4!" program.

Awards

Erich Vogt First Year Summer Research Experience (FYSRE) Award | 2024

Awarded to promising students in physics for a 4 month research placement.

Alberta Centennial Award and Alberta Premier's Citizenship Award | 2023

Awarded for outstanding community service. Value: \$2000

Calgary Flames Foundation Community Involvement Scholarship | 2023

Awarded for community involvement. Value: \$2005

Julia Turnbull Leadership Award for exceptional community service | 2023

Awarded for exceptional community service. Value: \$1000

Ted Rogers Entrance Scholarship | 2023

Awarded for academic achievement. Value: \$2000

Advocacy and Leadership

Highly Qualified Personnel (HQP) Advisory Committee Member

Arthur B McDonald
Astroparticle Physics Institute
2024 - Present

- Worked with members across Canada to develop opportunity lists for students and recent graduates.
- Shared McDonald Institute opportunities with eligible HQP in BC and Alberta.

Advisory Team Member

Child Rights Connect
2021 - 2023

- Provided guidance to UN delegations on communication strategies for high-level rights goals.
- Presented to governments and consulted on international initiatives to support the UN Convention on the Rights of the Child.

Experiences

- **"Quantum School for Young Students"** participant at the University of Waterloo and Institute for Quantum Computing.
- **"Introduction to Quantum Computing"** participant, an 8-month course on quantum computing using IBM's quantum infrastructure.
- **Scientific Computing with Python certification** (300 hours).